Best Country to Visit: A Composite Holiday Destination Index

Contents

Executive Summary

In this project I develops a data-driven composite indicator designed to identify and rank countries as optimal holiday destinations. Rather than relying on subjective travel guides or marketing materials, this index combines quantitative data across three critical dimensions:

1. Economic accessibility
2. Climate appeal
3. Safety risks in relation to disasters

The final variables used in the index synthesises data from 170 countries across 8 variables spanning cost of living, weather patterns, and natural disaster frequency. Through multivariate analysis and Principal Component Analysis these variables are condensed into interpretable sub-indices and aggregated into a single composite score (0–1), enabling direct country-to-country comparison.

Note that for this project I created a total of 4 unique indexes:

- Warm and cheap holiday index
- Cold and cheap holiday index
- Sweet spot holiday index, where I added preferred ranges for variables (e.g. cost range and temperature range) to better generalise realistic travel preferences
- A second sweet spot index combined with PCA

GitHub Link:

https://github.com/JdcStig/JamieDuffyCreagh_DAV_CA_Holiday

Section 1: Theoretical Framework

1.1 The Problem Statement: Why a Composite Holiday Index?

The Challenge

Holiday selection is a complex, multidimensional decision-making process. While choosing a travel destination is inherently subjective and driven by individual preferences like local cuisine, history, or culture these qualitative indicators are often difficult to quantify and standardize.

However, many travellers also need to consider:

Affordability, Personal safety and Weather conditions

This study prioritizes these three core dimensions that universally influence travel decisions:

- Cost: The baseline financial affordability of the destination
- Climate: The likelihood of pleasant and predictable weather conditions
- Safety: The absence of significant, disruptive natural hazards

---

The Solution: A Composite Indicator Approach

A composite indicator reduces multiple dimensions into a single synthetic score while preserving information about underlying drivers.

This approach:

- Aggregates complexity into an actionable metric (a single country ranking)
- Preserves interpretability through component sub-indices (Economy, Environment, Risk)
- Enables comparison across 170 countries using standardised, objective data

---

1.2 Data Sources

Data Sourcing and Methodological Limitations

A major challenge was gathering the data, since travel information is often fragmented across different sources.

To build this index, data was collected from two distinct types of sources:

Cost and Climate Data

- Sourced from open-access Kaggle datasets
- While these datasets provide immediate accessibility, they present limitations regarding like accuracy and standardization
- They were utilized as the most viable open-source option available

Safety and Natural Hazard Data

- Sourced from the Emergency Events Database (EM-DAT)
- Maintained by the Centre for Research on the Epidemiology of Disasters (CRED)
- Unlike the crowdsourced nature of the cost and weather data, EM-DAT is a highly accredited, peer-reviewed global database that provides reliable country-level records of historical disaster events and their socio-economic impacts

---

1.3 Assumptions and Notes

Due to limited data accessibility, much of the dataset did not extend beyond 2023. Some datasets, such as the cost index, only contained data for 2023.

Because of these limitations, the index was designed using data exclusively from 2023 to maintain consistency across all variables.

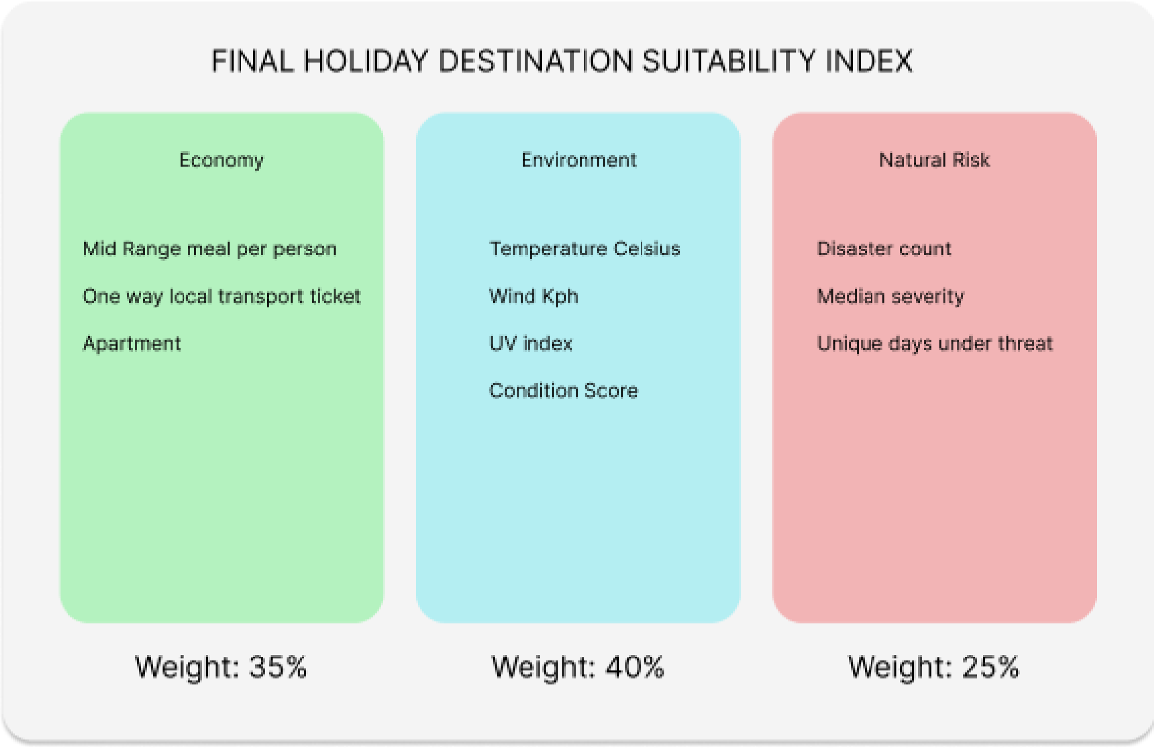
Additional assumptions:

- All costs were already converted to USD
- No currency conversion was required
- Since all data was limited to 2023, inflation adjustments were not considered

Finally , when merging all data sets if there were no records related to natural disasters I assumed that no disasters occurred in 2023 in relation to that country so I set them all to 0 rather than dropping countries that didn't appear.

1.4 Conceptual Framework: The Three-Pillar Model

The index is built on a three-pillar conceptual framework that captures the essential dimensions of holiday attractiveness based on the image below.



1.4.1 Pillar 1: Economic Accessibility

Rationale

The cost of a holiday is often one of the biggest factors influencing travel decisions. A destination's affordability determines whether a trip is financially realistic and how much money travellers can spend on activities, dining, and experiences.

The World Economic Forum Travel & Tourism Development Index 2024 rated price competitiveness at 4.5/7, suggesting it is a moderately important factor in tourism competitiveness, hence the weighting of 35%.

Source

Kaggle – Global Cost of Living Dataset

Definition

Economic accessibility measures the relative cost of living, transport, and food within a country.

Variables Used

Apartment Cost

Refers to the average price of a one-bedroom apartment in a city centre (USD).

One-Way Local Transport Ticket

Refers to the cost of a one-way local transport ticket per person (USD). This was chosen because it provides a more consistent measure of everyday transport costs.

Mid-Range Meal per Person

The original dataset measured the price of a three-course meal for two people at a mid-range restaurant (USD). In the analysis, I divided them by two to estimate the average cost per person.

All prices were already standardised in USD, so no currency conversion was required.

Theoretical Justification

- Universal relevance: Cost is an important consideration for travellers regardless of destination
- Quantifiable measurement: Cost-of-living datasets provide standardised country-level estimates that can be measured and compared consistently

---

Expected Relationship to Holiday Suitability

Lower costs are expected to increase holiday suitability, as travellers generally prefer more affordable destinations.

Therefore, a lower/more affordable cost index corresponds to a higher holiday suitability score.

1.4.2 Pillar 2: Environmental Appeal (Climate & Weather)

Rationale

Climate and weather conditions influence holiday satisfaction. Pleasant weather (moderate temperature and low rainfall) supports outdoor activities and comfort, while extreme or unsuitable conditions (excessive heat, cold, rain) degrade the experience.

I weighted this pillar at 40%, benchmarked against the George Washington University Adventure Tourism Development Index (ATDI), which allocates exactly 40% to environmental assets.

According to them, environmental assets carry the greatest weight because they have the unique power to offset a country's infrastructural or economic shortcomings.

Source

Kaggle – Global Weather Repository Dataset

Definition

Environmental appeal measures the climate conditions of a country and their attractiveness for tourism.

Note

According to the data source, data collection began on August 29, 2023.

The dataset contained daily weather recordings for each country. To ensure consistency across countries:

- Only the first 365 daily records per country were retained
- Data collection started from 2023-08-29
- The data was aggregated into a single representative row per country

---

Variables Used

Temperature

Measures the average temperature of a country. All values were already recorded in degrees, so no conversion was required.

Wind Speed (kph)

Measures the average wind speed in kilometres per hour.

UV Index

Measures the intensity of ultraviolet radiation.

During data cleaning, a value of 11 initially appeared causing me to think it was an error; however, UV index values of 11 or higher are officially classified as “extreme,” meaning the value was valid and retained.

Condition Category

The original dataset contained detailed weather descriptions such as “light rain shower” and “mist.”

Similar conditions were grouped into broader categories such like Rain or clear. This resulted in six overall categories.

These categories were then converted into numerical values by ranking them according to their perceived suitability for holidays (see image below).

Category	Score
Thunderstorm	0
Rain	1
Fog	2
Cloudy	3
Partly Cloudy	4
Clear	5

Theoretical Justification

- Evidence-based relevance: Weather conditions strongly influence tourism decisions and the types of activities available to travellers
- Objectively measurable: Weather datasets are easy to interpret and analyse

---

Expected Relationship to Holiday Suitability

Moderate temperatures, lower rainfall, and generally pleasant weather conditions are expected to increase holiday suitability.

Therefore, destinations with more favourable climate conditions are likely to achieve higher holiday suitability scores.

This depends on which index is being analysed. Again, I created:

- A cold and cheap holiday index
- A warm and cheap holiday index
- Two middle "sweet spot" holiday indices

1.4.3 Pillar 3: Safety & Risk (Natural Hazards)

Rationale

Safety is a major consideration when choosing a travel destination.

Natural disasters such like floods and earthquakes, can threaten visitor safety and disrupt travel through infrastructure damage, transport delays, flight cancellations, and emergency closures.

Definition

Safety risk measures both the frequency and severity of natural hazards within a country.

Variables Used

Disaster Count

The total number of natural disasters recorded in a country during 2023.

Median Severity

The median number of people affected per disaster within a country.

Unique Days Under Threat

Measures the number of distinct days during which a country experienced at least one natural disaster.

This accounts for situations where multiple disasters may occur at the same time.

Theoretical Justification

- Direct impact on safety: Natural disasters can result in injury, death, and displacement

- Economic disruption: Disasters can damage tourism infrastructure, including hotels, transport systems, and attractions
- Traveller perception: Perceived risk strongly influences destination choice, and media coverage of disasters may discourage tourism

---

Expected Relationship to Holiday Suitability

Lower natural disaster frequency and severity are expected to increase holiday suitability.

Countries experiencing fewer disasters are generally viewed as safer and more reliable travel destinations.

Section 2: Data Cleaning and Imputation

2.1 Pillar 1: Environmental Appeal (Climate & Weather)

Weather Data Cleaning and Preparation

When initially importing the weather dataset, I retained the following variables:

temperature_celsius, condition_text, wind_kph, humidity, sunrise, sunset

---

Data Quality Checks

Throughout the cleaning process, I first checked for any missing (NaN) or zero values within the dataset. After confirming that no significant missing values were present, I ensured that all variables had consistent and appropriate data types.

This validation step was important because in previous datasets I worked with, numerical variables were sometimes incorrectly stored as strings. Therefore, a general sanity check was performed to confirm data consistency.

I also validated that variables such as wind_kph, UV index, Humidity did not contain negative values, as these would not be statistically or physically possible.

Country Name Standardisation

One issue discovered during cleaning was that several country names were not recorded in English. For example, one record contained “火鸡,” which translates to “Turkey” in Chinese.

To ensure consistency across datasets and avoid issues during aggregation and merging, I created a script to standardise all country names into English. This was developed using online documentation and assistance from ChatGPT. After which I Standardise them all so that all countries start are in lower case to ensure consistency.

Aggregation of Daily Weather Records

During cleaning, I identified a total of 135,658 rows of weather data.

According to the dataset documentation, the data consisted of daily weather recordings for each country. To create a representative yearly weather profile for each country, I retained the first 365 daily records per country and aggregated them into a single country-level observation.

This allowed the dataset to better reflect long-term average weather conditions rather than short-term fluctuations.

Weather Condition Categorisation

The original dataset contained many detailed weather condition labels, including terms such as Drizzle, Shower, Mist and Light rain

Many of these categories were closely related or overlapping in meaning. To simplify the analysis, I grouped similar conditions into six broader weather categories.

To convert these categorical values into numerical data suitable for analysis, I assigned ranking scores to each category based on their perceived suitability for holidays. More desirable weather conditions received higher rankings, while less desirable conditions received lower rankings.

Feature Engineering

I also created a new variable related to sunlight exposure using the sunrise and sunset variables.

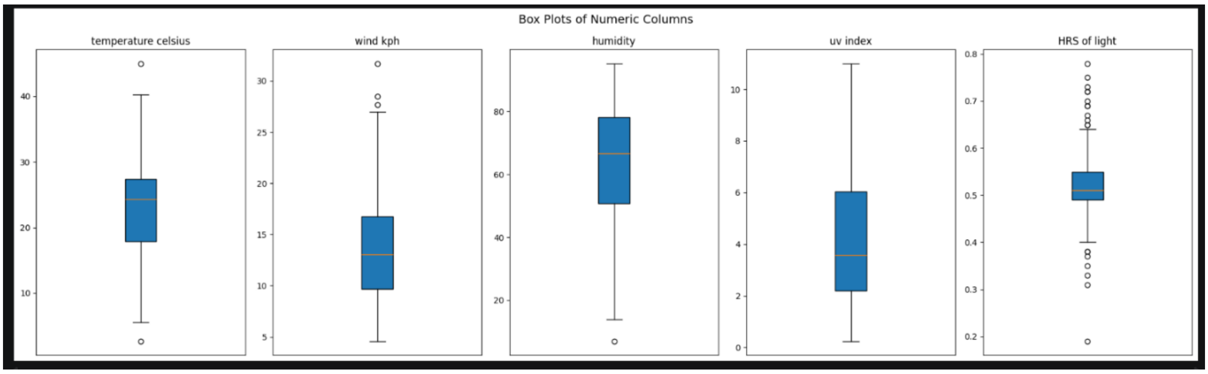
The sunrise and sunset times were converted into seconds, and the difference between them was calculated to estimate the total number of daylight hours each country received on average.

This additional variable was included as a potential indicator of environmental attractiveness.

Final Validation

After cleaning and transformation, I conducted range validation checks and Box plot analysis to identify any unrealistic values or potential outliers.

These final checks ensured that the cleaned dataset remain



Going through all outliers they all make sense in my opinion:

The outlier in temperature Celsius refers to Saudi Arabia with a temperature of 45 degrees. According to https://www.visitsaudi.com/en/stories/climate-and-seasons from June to September "the average hovering around 45 degrees C". So, this is accurate piece of data and doesn't make sense to remove it or replace with mean as it would not accurately represent its climate

The other outlier appears to be Canada with 5 degrees Celsius. According to https://www.studycanada.ca/english/climate.htm "Daytime summer temperatures can rise to 35°C and higher, while lows of -25°C are not uncommon in winter." so averaging that gives us around 10 degrees average so 5°C is a possible temperature so I will keep it.

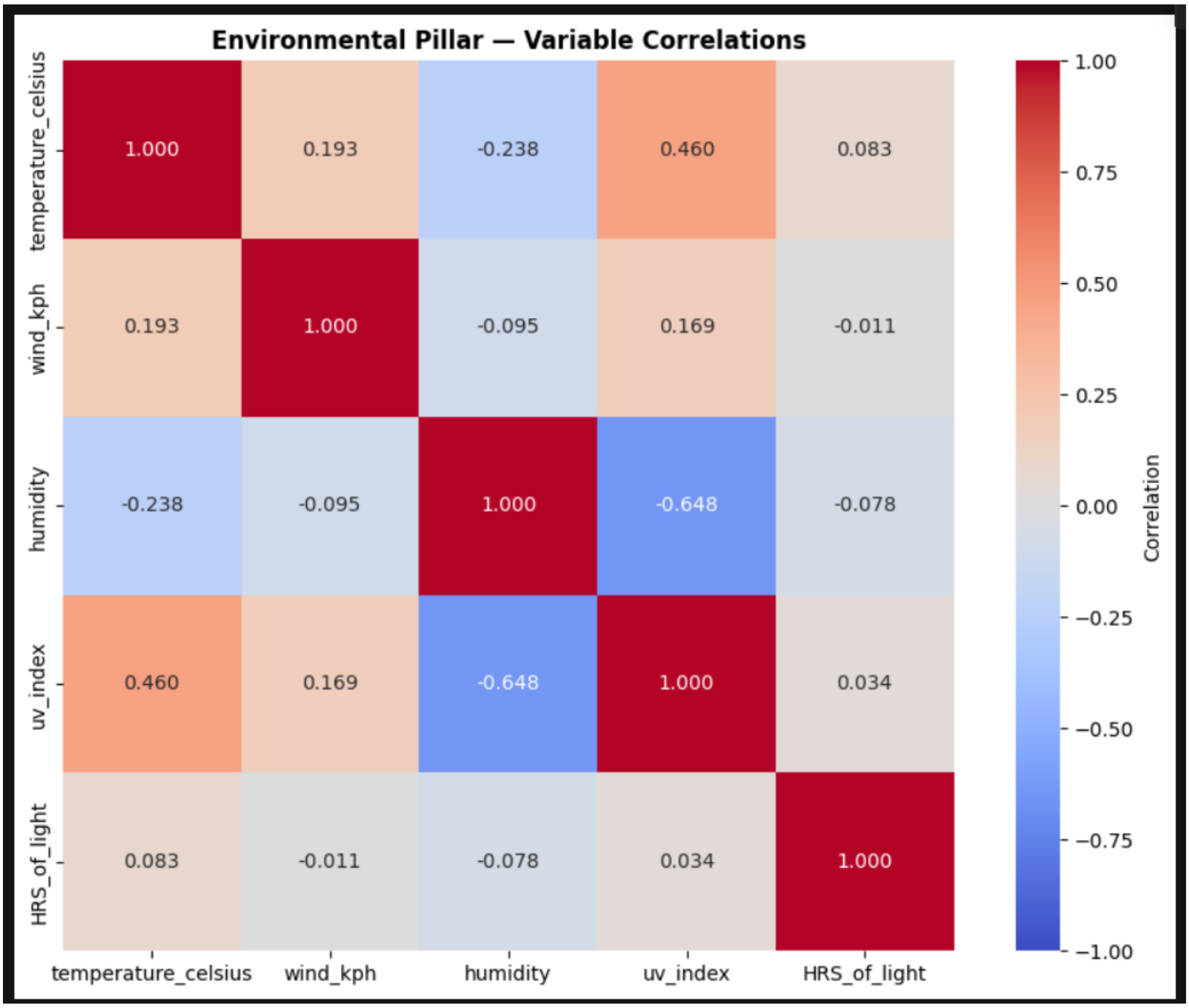
For Wind one of the values represents Iceland at 28km/h. According to https://rumbomundo.com/weather-in-iceland-temperature-wind-precipitation/#climate_iceland_temperature_wind_precipitation "gusts of more than 50 km/h (30 mph) can happen in any given month, and up to 90 km/h (56 mph) in the winter months. " These extreme values most likely dragged up the average wind kph to 28 but I will keep it in.

For Humidity Saudi Arabia was given a humidity level of 7 which, according to https://weatherspark.com/h/y/104018/2023/Historical-Weather-during-2023-in-

Riyadh-Saudi-Arabia it's safe to say this should be in the 30s at least so I went ahead and set this value to 30.

According to below Iceland had an average of 0.19 (4.56) hrs of sunlight on average and Sweden had roughly 18.72 hrs of light on average, these values seem to be heavily skewed due to seasonal conditions and seem to be invalid. However, I do eventually decide to drop this data due to high VIF and the fact that I realize it's a poor indicator for choosing the best holiday. Knowing that and due to not having enough time I'm not going to handle this data since I eventually drop it. I will go through this in greater detail later.

I then create a heat map to check for multicollinearity. The result of which is below:



The biggest concern here is that UV and humidity seem to be strangle negatively correlating, which makes sense, UV index increases when there's more sunlight, less sunlight often means cooler temperatures. Again, Ill mention how I remove values later.

2.2 Pillar 2: Economic Accessibility

The cleaning process for the cost-of-living dataset followed a similar structure to the weather dataset.

Initial Transformations

At the beginning of the process, I divided the Mid_Range_Meal_for_2 variable by 2 to estimate the average meal cost per person. In my opinion, this provides a more realistic and interpretable measure for individual travellers.

I also standardised all country names into title case format to ensure naming consistency across datasets and to avoid issues during merging and aggregation.

Handling Missing Values

I then identified and handled missing (NaN) values within the dataset.

Since multiple records existed for some countries, missing values were first imputed using the median of rows belonging to the same country. A country-specific median was chosen because it provides a more representative and realistic estimate than using a global value.

After this stage, some missing values remained (see image below).

```
Missing values before imputation:
Mid_range_meal_pp                  442
One_way_local_transport_ticket    1473
Apartment                         1341
----------------------------------------
Missing values after imputation:
Mid_range_meal_pp                    2
One_way_local_transport_ticket      12
Apartment                            5
----------------------------------------
```

Data Validation

Next, I performed data validation checks to ensure all variables had the correct and consistent data types.

Admittedly, this validation should ideally have been completed before handling missing values. However, the validation confirmed that all variables already had appropriate data types, so this did not affect the integrity of the cleaning process.

Removing Empty Records

I then checked whether any rows contained entirely missing values across all columns.

One row was found to contain no usable data and was therefore removed from the dataset.

For the remaining missing values, I applied global median imputation (see image below). This was necessary because some countries lacked enough observations to calculate a reliable country-specific median.

	country	Mid_range_meal_pp	One_way_local_transport_ticket	Apartment
3	American Samoa	3.28	NaN	88.18
6	Anguilla	85.00	NaN	800.00
7	Antigua And Barbuda	46.26	1.02	NaN
35	Chad	15.84	NaN	1103.64
39	Comoros	NaN	NaN	NaN
40	Cook Islands	24.86	NaN	161.54
72	Guinea-Bissau	7.51	NaN	341.22
124	Nauru	16.76	NaN	1676.56
132	North Korea	10.24	0.65	NaN
154	Sao Tome And Principe	20.78	NaN	207.74
183	Tonga	24.88	NaN	NaN
188	Turks And Caicos Islands	NaN	NaN	750.00
189	Tuvalu	5.20	NaN	113.43

```
Rows dropped (all numeric columns NaN): 1
Rows remaining: 203
----------------------------------------
Mid_range_meal_pp: 1 remaining NaN(s) filled with global median (16.3)
One_way_local_transport_ticket: 10 remaining NaN(s) filled with global median (0.79)
Apartment: 3 remaining NaN(s) filled with global median (393.905)
Missing values after fallback imputation:
Mid_range_meal_pp                 0
One_way_local_transport_ticket    0
Apartment                         0
```

Negative Value Checks

I also checked for negative values within the dataset.

Again, this step ideally should have occurred earlier in the cleaning process before imputation. However, no negative values were detected, meaning the order of operations did not impact the final results.

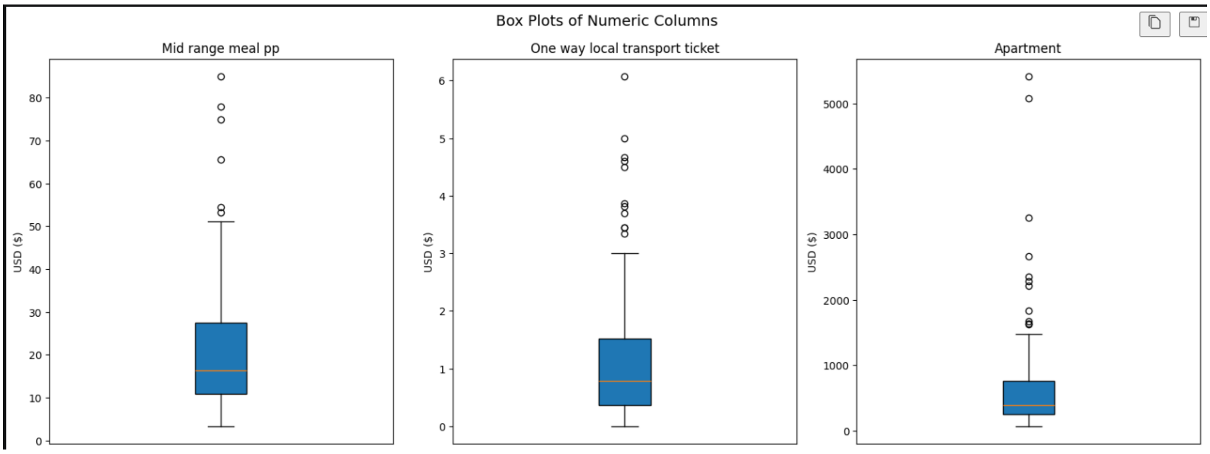
Range Validation and Outlier Analysis

After cleaning, I performed:

- Range validation
- Box plot analysis

to ensure all values were statistically plausible and economically realistic.

The box plot is shown below.



Treatment of Outliers

Although several significant outliers were present in the dataset, I decided to retain them rather than cap or replace them using global summary statistics such as the mean.

This decision was made because the dataset spans a wide range of global economies across regions including Asia, Europe and Africa

As a result, there is no single universal economic baseline that would accurately represent all countries.

Furthermore, due to limitations in the available data, I was unable to cluster countries into comparable regional economic groups. Replacing extreme values with a worldwide median would therefore distort the genuine economic conditions of many countries.

For this reason, the outliers were retained as legitimate representations of real global economic variation.

Potential Improvements

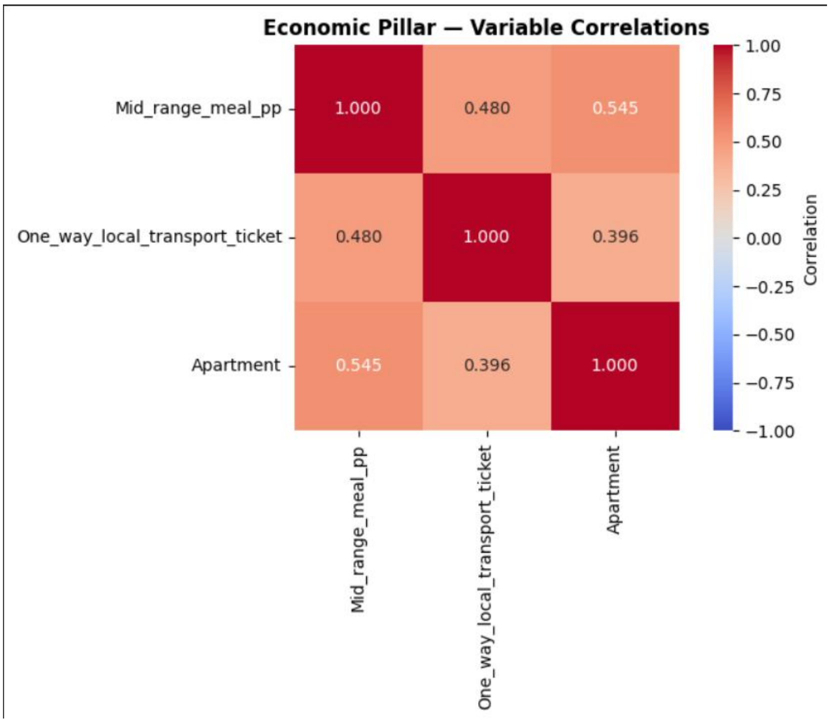
If a more detailed dataset were available, a more advanced imputation strategy could have been implemented.

For example, countries could be grouped by geographic or economic similarity (e.g. Asian countries, European countries, African countries). Missing values could then be imputed using regional medians or averages rather than global statistics, providing a more realistic reflection of expected economic conditions.

Multicollinearity Checks

Finally, I created a correlation heat map to check for multicollinearity between variables.

None of the relationships were considered strong enough to justify removing variables from the analysis, so all variables were retained for the final index construction.



2.3 Pillar 3: Safety & Risk (Natural Hazards)

The raw Natural Disasters dataset was first loaded and examined to understand its structure and overall quality.

A review of the unique disaster categories showed that the dataset contained 23 distinct disaster types across multiple countries. Each record included: Disaster event dates, Country information, Human impact metrics. This initial exploration helped establish an understanding of the dataset before cleaning and transformation began.

```
Unique Disaster Types:
<StringArray>
[             'Mass movement (wet)',                              'Flood',
                             'Rail',                 'Volcanic activity',
                       'Earthquake',                           'Epidemic',
                            'Storm',                              'Water',
            'Fire (Miscellaneous)',                                 'Air',
                             'Road', 'Miscellaneous accident (General)',
        'Collapse (Miscellaneous)',               'Extreme temperature',
           'Collapse (Industrial)',             'Explosion (Industrial)',
               'Fire (Industrial)',          'Explosion (Miscellaneous)',
                         'Wildfire',                           'Gas leak',
      'Glacial lake outburst flood',    'Industrial accident (General)',
                          'Drought']
Length: 23, dtype: str
```

Column Selection

To focus the analysis, only the most relevant variables were retained: Disaster Type, Country, Start Month, Start Day, End Month, End Day, Total Affected, No. Injured, No. Affected and No. Homeless.

Data Quality Audit

I then checked for missing values and NaN values across all columns.

```
Missing value counts per column:
----------------------------------------
Country:              0 NaN  |  0 blank
Disaster Type:        0 NaN  |  0 blank
Start Month:          5 NaN  |  0 blank
Start Day:            74 NaN  |  0 blank
End Month:            6 NaN  |  0 blank
End Day:              66 NaN  |  0 blank
No. Injured:          371 NaN  |  0 blank
No. Affected:         300 NaN  |  0 blank
No. Homeless:         592 NaN  |  0 blank
Total Affected:       168 NaN  |  0 blank
```

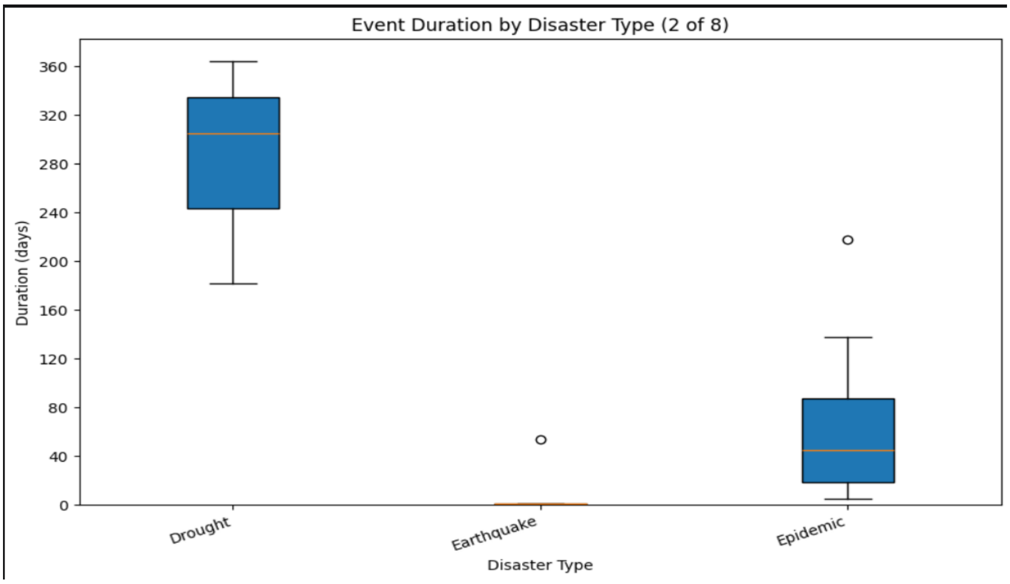
Event Duration Analysis

To create a reliable basis for imputing missing dates, I calculated disaster using the difference between event start and end dates.

I applied 2 rules during this process:

- Same-day disasters were treated as lasting 1 day
- If an event's end date appeared earlier than the start date within the calendar year, the end date was capped on 31 December 2023 rather than rolling into the following year.

I then visualised the data using box plots.



Outlier Investigation and Validation

Several duration outliers were identified and manually reviewed.

Iceland Earthquake (54 Days)

One notable outlier was the 2023 Iceland earthquake sequence, which appeared to last from October 25 to December 18.

Initial inspection suggested this duration was unrealistic for a single earthquake event. Further background research revealed that while seismic activity continued across this period, the primary earthquake (magnitude 4.6) occurred on October 25 and lasted only a short period of time.

Since the dataset was intended to capture the main disaster event rather than the extended aftershock sequence, the duration was adjusted to 1 day.

Somalia Cholera Epidemic (~3 Months)

Another significant outlier was a cholera epidemic recorded in Somalia lasting approximately three months.

After reviewing supporting information, this duration was confirmed as a legitimate event and was retained within the dataset.

Overall, identified outliers were retained where they reflected genuine real-world disasters.

Missing Date Imputation

Missing date values were imputed using disaster-type-specific median durations.

Partial Date Imputation

Where one side of the event date was available:

- If the start date was missing but the end date was known: start date = end date – median duration
- If the end date was missing but the start date was known: end date = start date + median duration

Fully Missing Date Imputation

For records where both the month and day were missing:

- The start month was set to the modal month for that disaster type
- The start day was fixed to the 15th of the month
- The end date was then estimated using the median disaster duration

This approach helped preserve the typical temporal behaviour of each disaster type while minimising unrealistic assumptions.

Impact Field Imputation

For records where Total Affected was missing, the value was calculated using No. Injured No. Affected No. Homeless

Missing values within these component fields were treated as 0 during calculation. As I assumed if no record of a disaster was made for a country then it did not suffer any disasters in 2023.

This method aligned with the EM-DAT documentation, which defines Total Affected as the sum of these categories.

Final Validation

Following all cleaning and imputation steps, the dataset was re-validated to ensure no remaining missing values existed.

The final dataset contained zero NaN values across all retained columns.

Feature Engineering – Disaster Categorisation

To simplify analysis and reduce dimensionality, the original 23 disaster types were grouped into 4 broader categories: Natural Disasters, Industrial Disasters, Transport Disasters and Miscellaneous / Other Disasters

This hierarchical grouping made the dataset easier to analyse while still preserving meaningful distinctions between disaster categories.

I then created the following

Disaster_Count: Total number of individual disaster events a country suffered from during 2023

Median_Severity: Typical severity across all events (0–1 scale) , calculated from the total affected field. It was a crude scale I manually choose to set. It's like so :

None': 0.00, 0 affected

Small-scale': 0.25, # 1 – 999

Medium-scale': 0.50, # 1 000 – 9 999

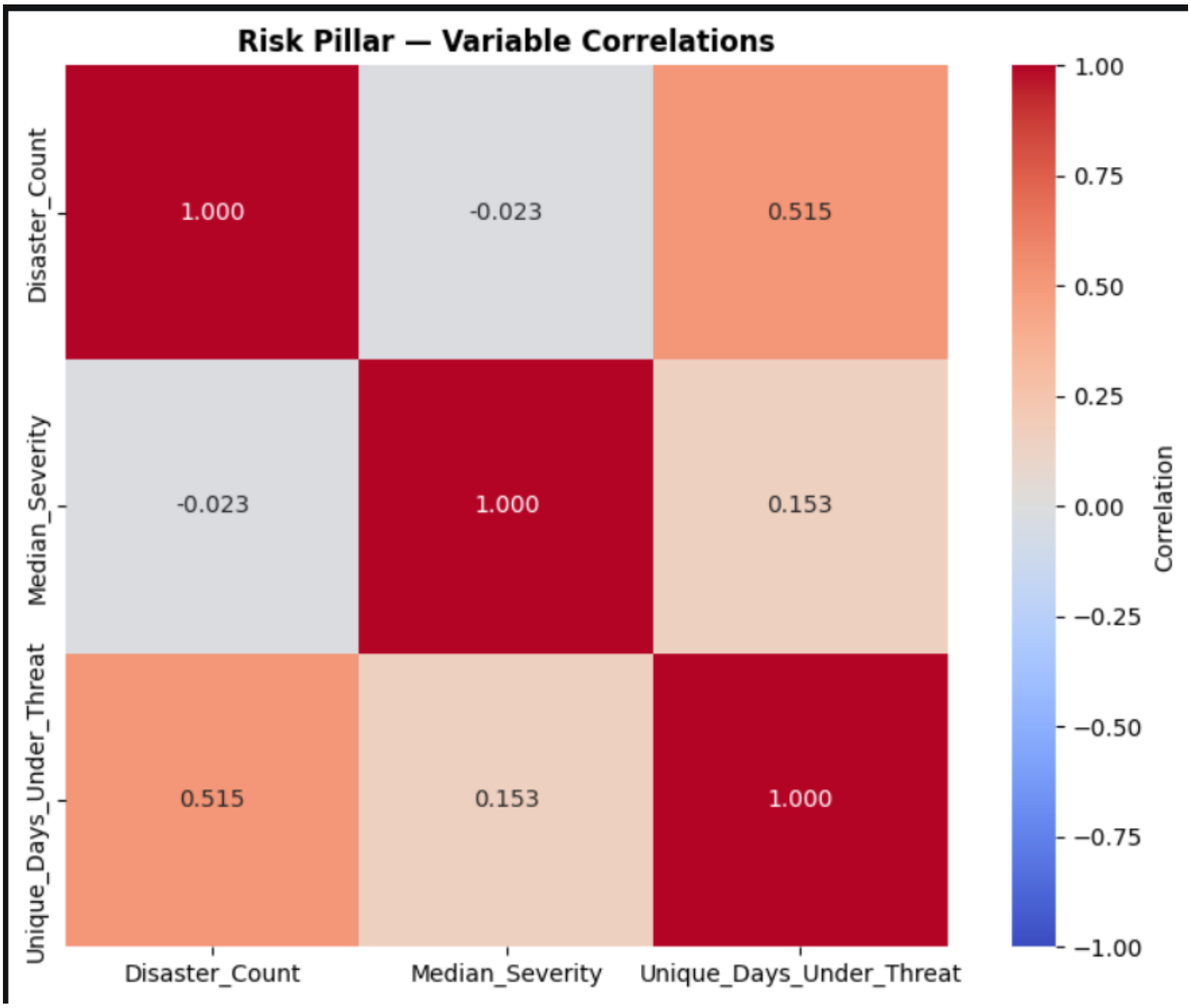
Large-scale': 1.00, # 10 000+

Unique_Days_Under_Threat: Shows how many actual unique days that country suffered from a natural disaster, as in some cases countries suffered from multiple disasters at the same time.

Multicollinearity Checks

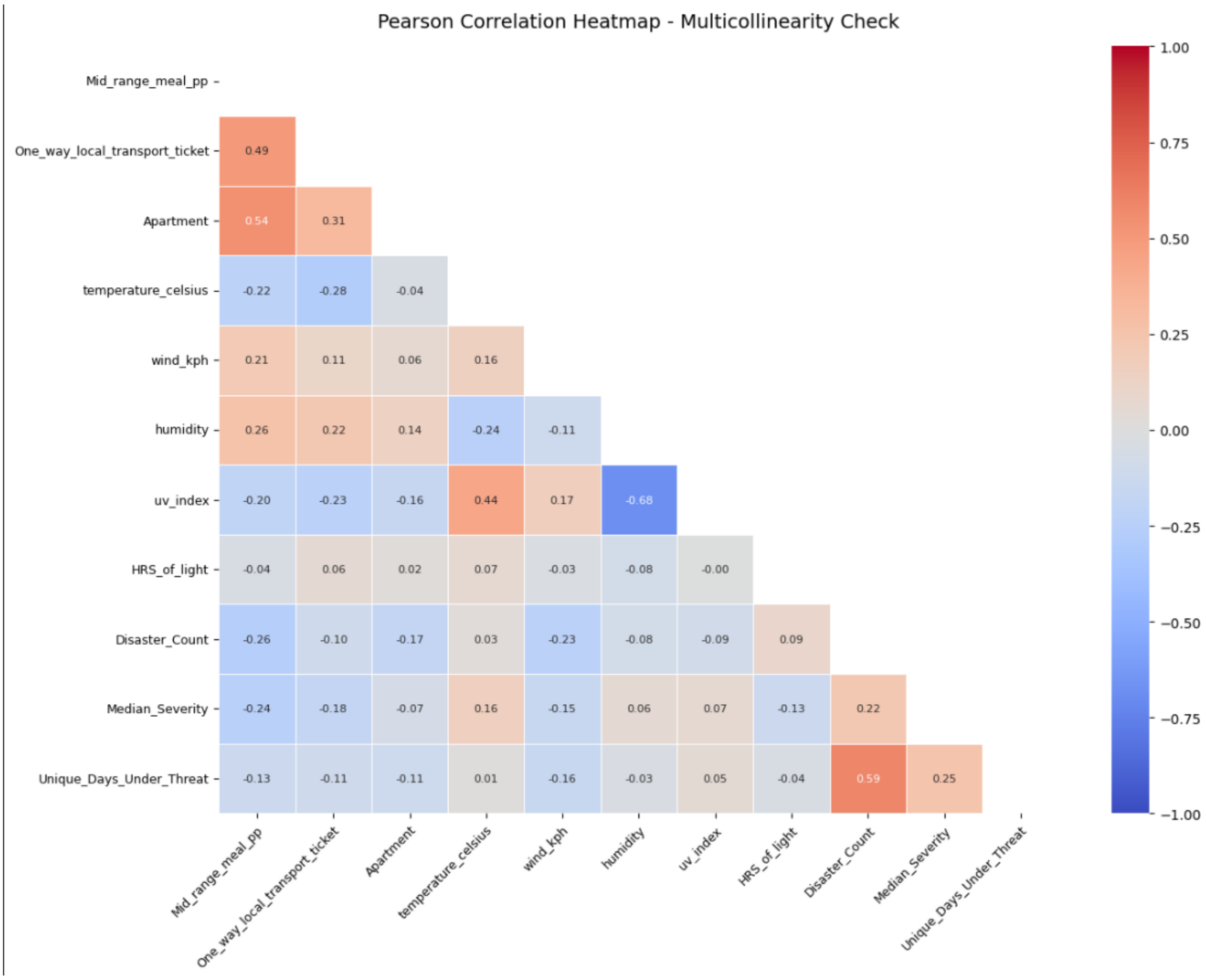
Finally, I created a correlation heat map to check for multicollinearity between variables.

None of the relationships were considered strong enough to justify removing variables from the analysis, so all variables were retained for the final index construction.



2.4 Handling Multicollinearity and dropping

Originally, I had a collinearity of -0.68 between UV and humidity. I originally stated I was going to keep both as I couldn't compare which was more important in relation to the dependent variable as I'm creating said dependent variable. I was also planning to let PCA handle the multicollinearity later on. However, during the combining pillars process I re did the heat map and conducted VIF.




```
2) VIF table (VIF > 10 is typically high)
HRS_of_light                          :   27.54
humidity                              :   18.05
temperature_celsius                   :   16.31
wind_kph                              :    9.04
uv_index                              :    7.00
Mid_range_meal_pp                     :    6.80
One_way_local_transport_ticket        :    3.00
Disaster_Count                        :    2.61
Apartment                             :    2.59
Median_Severity                       :    2.13
Unique_Days_Under_Threat              :    2.00
```

At this point I realized that the correlation between the 2 was too strong to ignore. Ultimately, I decided to drop humidity due to the fact it had a higher VIF and also because I feel temperature is an easier variable to explain and understand.

Also based on the VIF values I decided to also drop HRS_of_light.

Section 3: PCA, Clustering, Weighting and Normalisation

3.1 PCA

Next, I conducted a Principal Component Analysis.

In my opinion, this was a necessary step because the index combines multiple correlated indicators across weather, cost, and disaster severity. PCA helps reduce these correlated variables into a smaller number of orthogonal components that capture the dominant patterns within the data.

This dimensionality reduction provides several advantages:

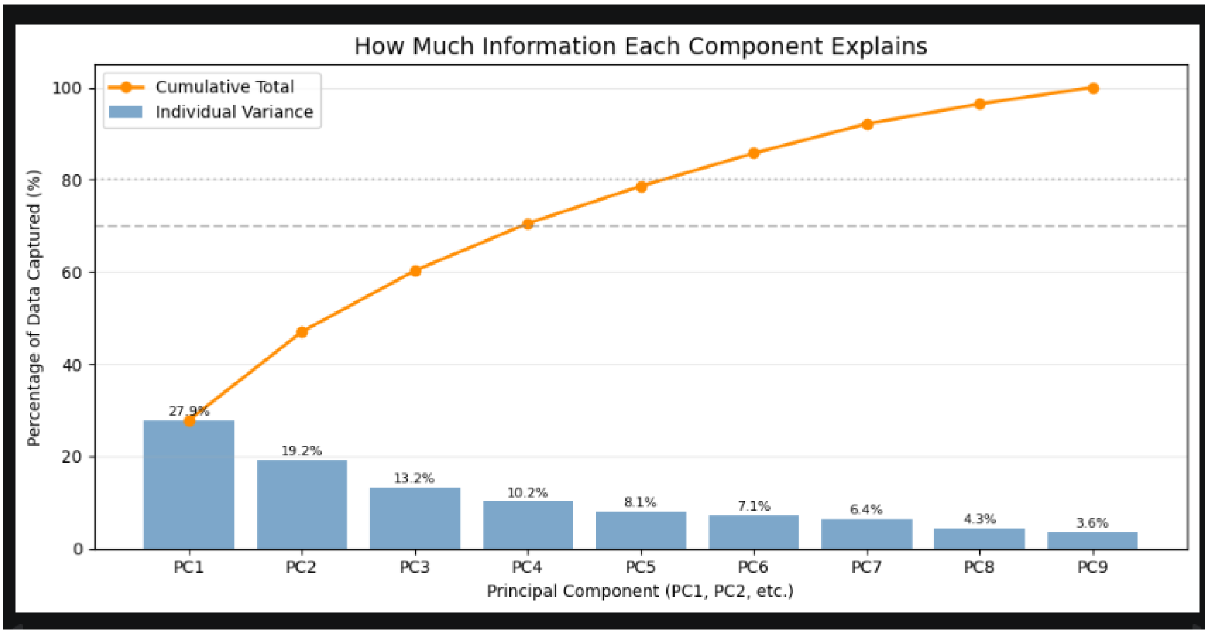
- Reduces multicollinearity between variables
- Minimises noise from less informative variables
- Produces more compact and stable weighted scores
- Prevents any single variable from disproportionately dominating the index

For example, without PCA, a variable such as apartment price could potentially become an overly influential factor within the final holiday suitability score.

Although PCA reduces interpretability to some extent, I believe the trade-off is worthwhile because it creates a more balanced and statistically reliable composite indicator.

To determine the number of principal components to retain, I used the cumulative explained variance method with an 80% variance threshold. A scree plot was then analysed to identify how many components were required to meet this threshold.

The results showed that 6 principal components needed to be retained to explain approximately 80% of the total variance within the dataset.



3.2 Clustering

The goal of my project is to build a **Best Country to Visit ranking index (0.00–1.00)** for travel decisions, not to force countries into a small number of hard groups. From an index-design perspective, clustering is optional and can reduce interpretability for my objective.

1. **Research objective mismatch**

Clustering answers: "Which countries are similar enough to group together?"

My goal is to answers: "How good is each country based on a travel score?"

PCA is directly aligned with ranking while clustering is primarily a segmentation tool.

2. **Weak natural separation in the data**

Using the same 6-principal-component representation retained for the index which explained approximately 86% of the total variance, I then evaluated how effectively K-means clustering could group countries into meaningful clusters.

To determine the optimal number of clusters, I calculated the **silhouette score** for k values ranging from 2 to 8.

The silhouette score measures how well each observation fits within its assigned cluster compared to other clusters. It evaluates both:

- **Cluster cohesion** - how similar observations are within the same cluster
- **Cluster separation** - how distinct each cluster is from neighbouring clusters

Silhouette values range from:

- **1** meaning Strong clustering, where observations are well matched to their own cluster and clearly separated from others
- **0** meaning Weak clustering, where observations lie close to cluster boundaries
- **Negative values** meaning Poor clustering, where observations may have been assigned to the wrong cluster

The resulting silhouette scores were relatively low, ranging between approximately 0.23 and 0.28 across the tested k values.

In practice, these results suggest that the dataset contains only weak cluster structure rather than strongly separated country groupings. Based on these results as well as the nature of clustering not aligning with my goal I decided to exclude it from my project.

3. **Information loss from hard labels**

Cluster labels collapse continuous variation into a few categories. Two countries with meaningfully different costs or risk profiles can end up in the same bucket, which is less useful than a ranked index when making destination comparisons.

3.3 Weighting

Throughout the weighting process, the final pillar weights were determined using a combination of personal judgement, supporting literature, and benchmarking against established tourism and development indices.

The final weighting structure was:

- **Environment:** 40%
- **Economy:** 35%
- **Risk:** 25%

This weighting system was designed to reflect the relative importance of each factor in influencing holiday decision-making while still maintaining a balanced and realistic composite index.

Environment – 40%

The **Environment** pillar was assigned the highest weighting because climate conditions, weather comfort, and seasonal stability are often the primary *pull factors* that motivate travel in the first place.

Pleasant weather directly influences Outdoor activities, Tourism satisfaction, Comfort levels and Overall destination attractiveness

This weighting was benchmarked against the **George Washington University Adventure Tourism Development Index (ATDI)**, which allocates 40% of its total weighting to environmental and resource-related factors, including natural assets, climate resilience, and tourism operating conditions.

The ATDI specifically highlights that strong environmental assets can compensate for weaknesses in other areas such as infrastructure or economic limitations. In practice, travellers are often willing to tolerate higher costs or reduced infrastructure if a destination offers highly desirable environmental conditions.

Additional supporting evidence also suggests that weather plays a major role in tourism decision-making. According to Medium, over 80% of travellers consider weather to be a significant factor when planning trips.

Based on these combined findings, I concluded that the environmental pillar justified the largest weighting within the index.

Sources

- Medium – Why Weather is Crucial for Travel
- George Washington University Adventure Tourism Development Index (ATDI)

Economy – 35%

The **Economy** pillar received the second-highest weighting because affordability acts as the immediate practical constraint on whether travel is financially feasible.

Even if a destination has attractive weather and strong tourism appeal, excessively high costs may discourage travellers or redirect them toward more affordable alternatives.

This weighting aligns with the World Economic Forum Travel & Tourism Development Index 2024, which identifies **Price Competitiveness** as a key enabling condition within tourism development. The index evaluates factors such as Purchasing Power Parity, Accommodation pricing and Consumer affordability

The WEF framework recognises that tourism demand is heavily influenced by relative costs and traveller spending power.

Additional evidence from Statista supports this relationship. Their findings indicate that approximately 60% of European travellers prioritise “good value for money” when selecting destinations (see image at end). Interestingly, the same survey showed that warm weather ranked lower than value for money, highlighting that affordability can outweigh climate preferences for many travellers.

This variation in preferences was one of the reasons I decided to create multiple versions of the index:

Based on both supporting literature and practical travel behaviour, the economy pillar was assigned a weighting of 35%.

Sources

- WEF Travel & Tourism Development Index 2024
- Statista – Influencing Factors on European Traveller Destination Decisions

---

Risk – 25%

Although safety is an important consideration in tourism, the **Risk** pillar received the lowest weighting because it functions more as a baseline constraint rather than an active travel motivator.

In many cases, travellers first choose destinations based on attractiveness, climate, or affordability, and only later evaluate whether the destination is sufficiently safe.

This weighting decision was supported by research published in *Progress in Tourism Management* by Karl (2020), which examined how travellers perceive and prioritise different forms of risk.

The study found that travellers are most sensitive to Health crises, Terrorism and Political instability

while natural hazards and disaster risks generally occupy a lower level of concern during early-stage destination planning.

This suggests that disaster risk often behaves as a filtering mechanism rather than a primary motivator for destination choice.

Additional evidence from Statista also showed that safety ranked below both weather and value for money when travellers evaluated destinations.

Despite this, I did not want safety to receive too small a weighting because natural disasters can still severely impact Traveller safety, Infrastructure reliability, Flight schedules and Tourism operations

As a result, the risk pillar was ultimately assigned a balanced weighting of 25%, ensuring that safety remained an important component of the index without overshadowing the core attractiveness factors.

Sources

- Karl (2020) – Risk Perception and Travel Decision-Making
- Statista – Influencing Factors on European Traveller Destination Decisions

3.4 Normalisation

For the normalisation process, I used **Min-Max normalisation**.

This method scales each indicator to a standard range between **0 and 1**, producing a comparable decimal score for every variable regardless of its original unit or scale.

This approach ensured that variables such as:

- Temperature
- Apartment prices
- Disaster frequency
- Wind speed

could all be directly compared within the same index framework.

Inverted Indicators

For indicators where **lower values represented better outcomes**, the normalised score was inverted using 1 – normalised value

This was primarily applied to variables such as:

- Cost indicators
- Disaster frequency
- Risk-related measures

where lower raw values should correspond to higher holiday suitability scores.

Index-Specific Adjustments

The direction of normalisation varied depending on the specific holiday index being created.

Warm and Cheap Holiday Index

In the **Warm and Cheap Holiday** index:

- Higher temperatures received higher scores
- Lower temperatures received lower scores

This reflected the assumption that warmer climates are generally more attractive within this version of the index.

Cold and Cheap Holiday Index

In the **Cold and Cheap Holiday** index, the temperature scoring was inverted:

- Lower temperatures received higher scores
- Higher temperatures received lower scores

This adjustment acknowledged that not all travellers prefer warm destinations, and that colder climates may also be desirable depending on travel preferences.

Importance of Normalisation

Applying Min-Max normalisation ensured that:

- No variable dominated the index purely because of scale differences
- All indicators contributed proportionally to the final score
- Variables with different units could be aggregated into a single composite index

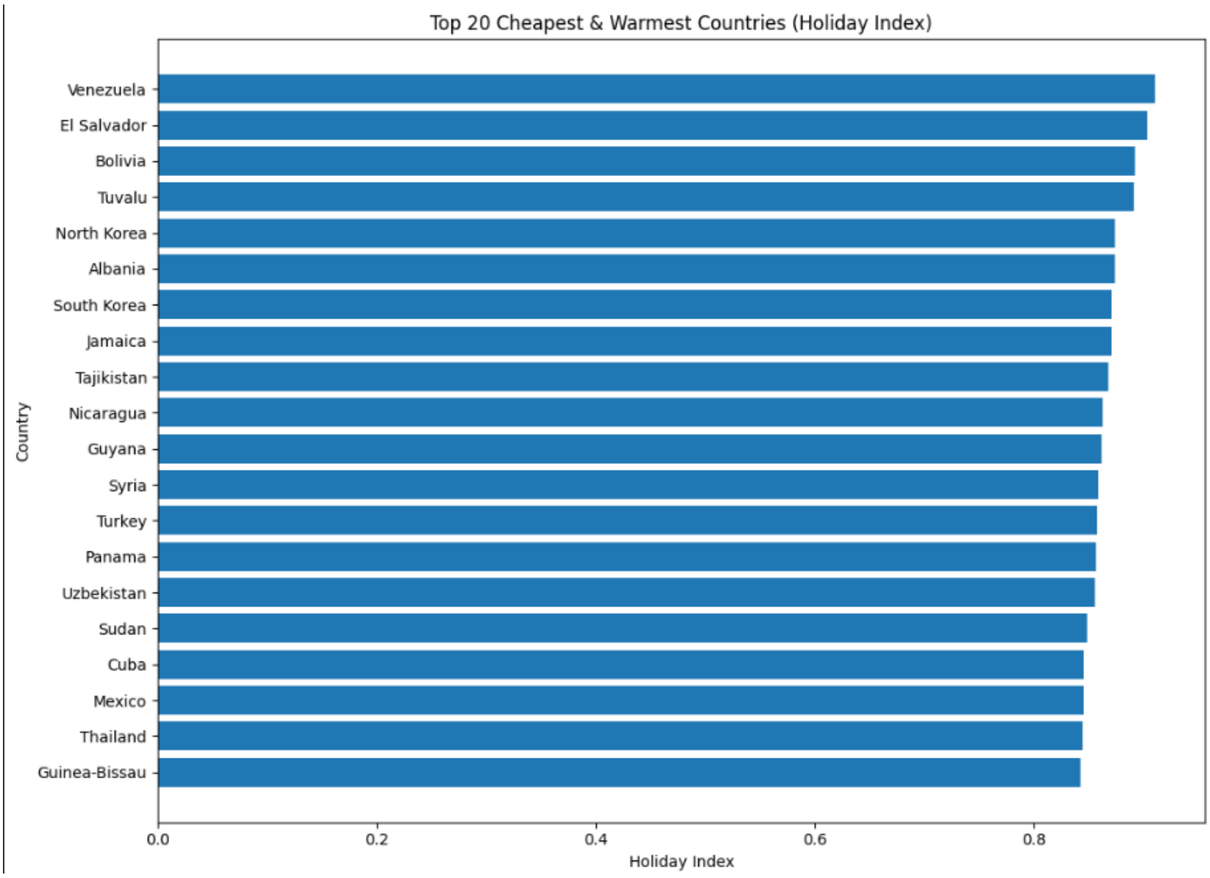
This created a more balanced and interpretable holiday suitability framework.

Section 4: Results

Note that for both warm cold and sweet spot (first one) Indexes I did not use PCA on them.

4.1 Cheapest & Warmest Country to Visit

The results were as follows:



This chart suggests that Venezuela, El Salvador, and Bolivia achieved the highest scores on the cheap-and-warm holiday index. Based on the weighting structure used in the model, these countries offered the strongest combined balance of affordability, warmer climate conditions, and relatively lower disaster-related risk within the dataset.

Several countries appearing in the top rankings align reasonably well with real-world travel expectations. Destinations such as Jamaica, Panama, and Thailand are widely recognised for their warm climates and comparatively affordable travel costs when compared with more expensive tourism markets in Europe or North America.

However, some results appear less intuitive. South Korea ranked highly despite not typically being associated with “cheap warm holidays.” South Korea experiences a temperate climate with distinct cold winters and relatively mild summers, meaning it does not fit as a typical warm holiday. In addition, while South Korea can be cheaper than destinations such as Japan or Western Europe, it is still generally considered more expensive than many other Asian tourism markets, particularly for accommodation and dining in major cities such as Seoul.

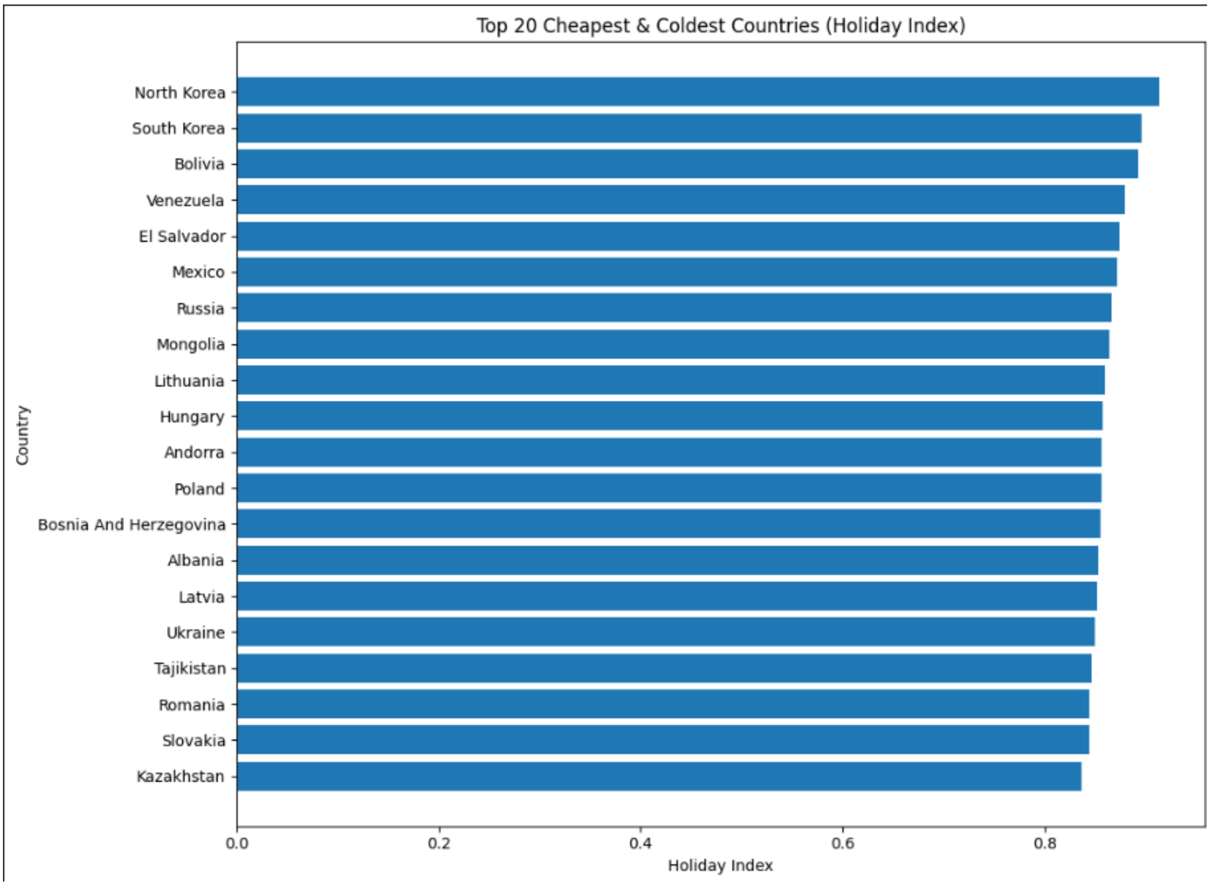
From a disaster-risk perspective, South Korea is also exposed to several natural hazards, including typhoons, flooding, and regional seismic activity. As a result, its position within the top-ranked countries is likely due to:

- The use of annual average temperatures
- The weighting balance between economy, climate, and safety pillars

Overall, the index performs reasonably well as a broad indicator combining affordability, climate, and disaster-related safety. Many of the highly ranked countries match commonly perceived budget-friendly warm destinations. However, certain rankings demonstrate that the model still simplifies complex tourism behaviour and poor rankings.

4.2 Cheapest & Cold Country to Visit

The results were as follows:



According to the chart, North Korea, South Korea, and Bolivia achieved the highest scores on the cheap-and-cold holiday index, suggesting that they offered the strongest overall combination of lower travel costs, colder climate conditions, and lower disaster-related risk within the dataset.

Several countries appearing in the rankings align reasonably well with real-world expectations. Destinations such as Russia, Mongolia, and Poland are widely recognised for colder climates and comparatively affordable travel costs when compared with Western Europe or North America. Their inclusion therefore supports the overall validity of the index.

However, some results appear less intuitive. Countries such as Mexico, El Salvador, and Venezuela are generally perceived as warm-weather destinations rather than cold tourism markets. Their high rankings are likely influenced by mountainous or high-altitude regions lowering average recorded temperatures within the dataset, even though the countries themselves are not commonly associated with cold-climate tourism overall.

South Korea presents a more complex case. While the country does experience genuinely cold winters, particularly in northern and inland regions, it is not typically considered a low-cost destination compared with many parts of Asia. Accommodation, dining, and tourism costs in cities such as Seoul are generally significantly higher than destinations across Southeast Asia. Travel cost comparisons consistently classify South Korea as a mid-range or relatively expensive Asian destination rather than a budget travel market. Also due to the fact its so highly rated in both leads to suspicion on how well the index performs.

In addition, South Korea is exposed to several natural hazards, including typhoons, flooding, and regional seismic activity, making its high ranking less intuitive from a disaster-risk perspective.

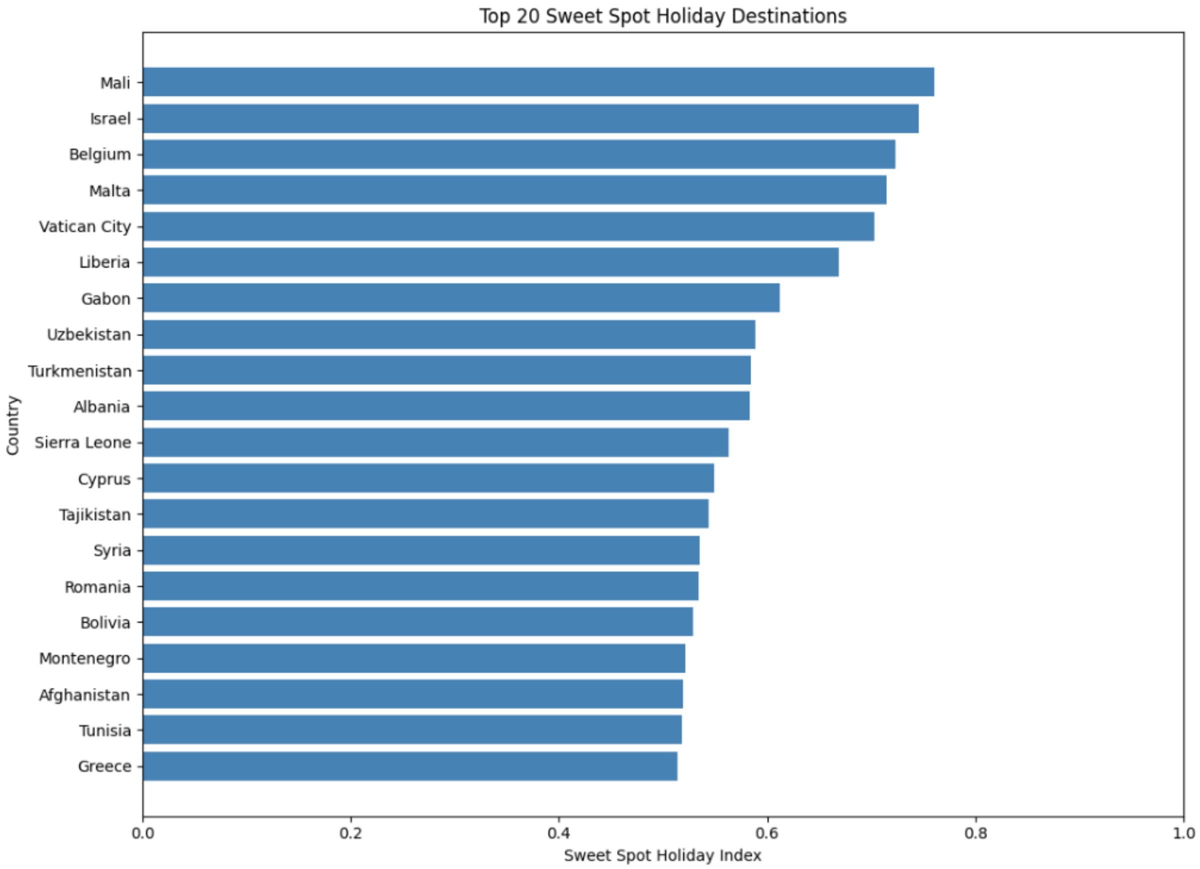
Overall, the index performs reasonably well as a broad statistical combination of affordability, colder climate conditions, and safety. Many of the highly ranked countries align with expected cold-weather budget destinations. However, some unexpected rankings demonstrate the limitations of composite indices, particularly when reducing complex tourism behaviour into a single quantitative score.

4.3 Sweet spot holiday (No PCA)

Note that for this index I leveraged ChatGPT quite a bit as I was unsure on how I would be able to achieve and set a sweet spot range. Throughout the process I tried playing around with ranges and sweet spots and eventually I arrived at the following:

Pillar	Indicator	Scoring Type	Preferred Zone / Direction	Bounds
Environment	`temperature_celsius`	target	23 - 26 degC	worst: 8 to 33
Environment	`wind_kph`	target	7 - 14 kph	worst: 26 and up
Environment	`uv_index`	target	4.0 - 5.5	worst: 3 and bellow to 7 and up
Environment	`condition_score`	high	higher is better	0 to 5
Economy	`Mid_range_meal_pp`	target	$8 - $70	no worst 100 and up
Economy	`One_way_local_transport_ticket`	target	$0.40 - $15	no worst worst: $0 to $30
Economy	`Apartment`	target	$250 - $1200 /mo	No worst: $2,500
Risk	`Disaster_Count`	low	lower is better	0 to 15
Risk	`Median_Severity`	low	lower is better	0 to 1
Risk	`Unique_Days_Under_Threat`	low	lower is better	0 to 365

With these range my index gave me the following results:



According to the chart, Mali, Israel, and Belgium achieved the highest scores on the Sweet Spot Holiday Index, suggesting that they offered the strongest overall balance between affordability, climate conditions, and disaster-related safety within the dataset.

Unlike the previous warm-only or cold-only indices, this “sweet spot” index was designed to reward countries that fall within more moderate and balanced ranges rather than favouring temperature extremes.

Several countries in the rankings fit this expectation reasonably well. Mediterranean destinations such as Malta, Greece, Cyprus, Montenegro, and Albania are commonly associated with mild climates, relatively stable weather conditions, and lower travel costs compared with many Western European tourism markets. Their appearance within the top rankings therefore aligns well with the intended purpose of the index.

Some of the higher-ranked countries may initially appear unexpected, particularly countries such as Mali, Liberia, Gabon, and Sierra Leone. However, within the context of this index, their rankings can be explained statistically through the variables included in the model.

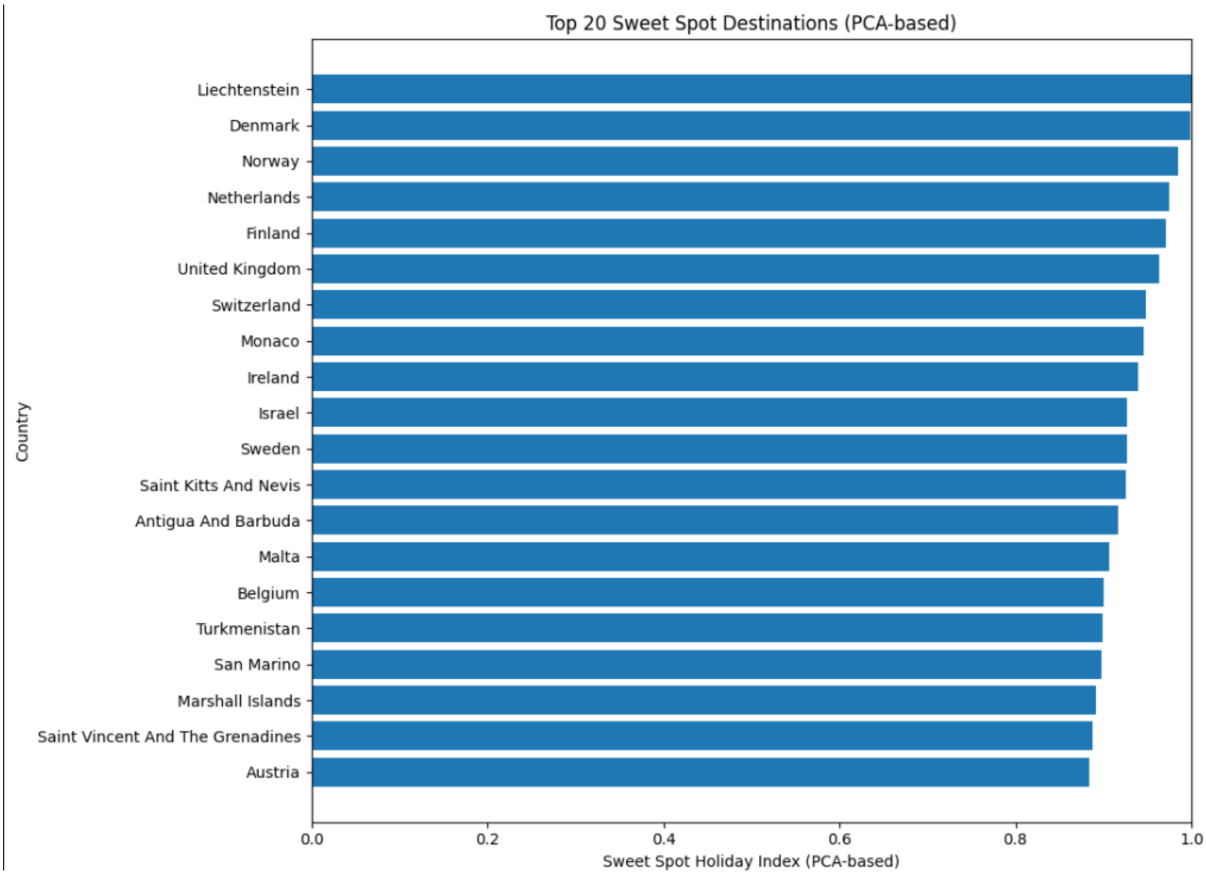
Similarly, Israel and Belgium performed strongly because they achieved relatively balanced scores across all three pillars rather than excelling in only one area. Although they are not among the cheapest destinations globally, they likely benefited from moderate climate conditions and comparatively lower disaster-related risk scores

Overall, the Sweet Spot Holiday Index appears more balanced and realistic than the warm-only and cold-only indices because it avoids rewarding extreme temperature conditions. Many of the top-ranked countries align reasonably well with destinations that offer moderate weather, affordability, and relatively lower disaster exposure.

At the same time, the rankings highlight an important limitation of composite indices: the final scores are highly dependent on the selected variables, weighting structure, and normalisation process.

4.4 Sweet spot holiday (With PCA)

Using the exact same ranges the results were as follows:



According to the chart, the **Sweet Spot Holiday Index combined with PCA** yields the most robust, objective, and realistic rankings generated in this project so far. By using PCA to condense the variables this model eliminates ad-hoc weighting and accounts for natural correlations between cost, weather patterns, and disaster risks.

Like the baseline sweet spot index, this advanced model was specifically calibrated to penalize temperature extremes and high consumer costs, instead rewarding destinations that consistently fall into preferred "sweet spot" ranges. Traditional Mediterranean favourites such as Malta, Greece, Cyprus, Montenegro, and Albania continue to perform exceptionally well. This outcome is entirely expected, as their moderate seasonal climates, stable environmental baselines, and relative affordability compared to premium Western European hubs.

Overall, the **Sweet Spot Index combined with PCA** stands out as the definitive best index developed so far in this project. It successfully moves past the linear constraints of the warm-only and cold-only models, while introducing a mathematically rigorous normalization and weighting process that respects how travel variables interact in the real world.

According to country to visit rankings by Lost in Travel https://lostintravel.blog/2024/01/20/every-country-i-visited-in-2023-ranked/:

Liechtenstein was ranked 14th

Denmark 6th

Norway (did not appear)

Netherlands 13th

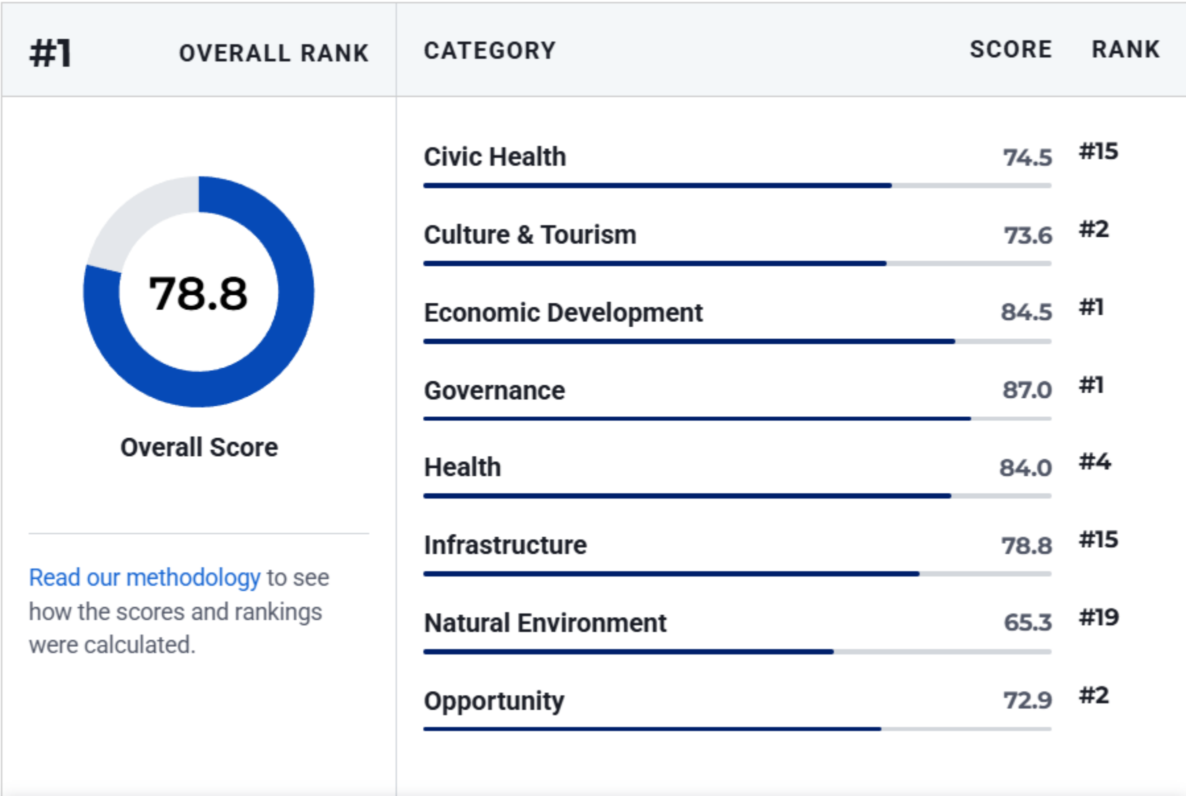
Finland (did not appear)

This site is based on an individual opinion, and he only ranked a total of 15 places he visited so some places did not appear in his ranking (like Norway) because he did not visit them.

One interesting site I came across is US news where they provided top 100 rankings of countries to visit in 2026. Comparing their rankings to model I got the following. My indexes results are on the left and US news on the right

1st – Liechtenstein , did not appear
2nd – Denmark , also ranked it as 2nd
3rd – Norway , ranked them 6th
4th – Netherlands , ranked them 5th
5th - Finland , ranked them 8th
6th - United Kingdom , ranked them 7th
7th - Switzerland , ranked them 1st
8th - Monaco , did not appear
9th - Ireland , ranked them 13th
10th - Israel , did not appear
11th - Sweden , ranked them 3rd
12th - Saint Kitts And Nevis , did not appear
13th - Antigua And Barbuda , did not appear
14th - Malta , ranked them 33rd
15th - Belgium , ranked them 11th
16th - Turkmenistan , did not appear
17th - San Marino , did not appear
18th - Marshall Islands , did not appear
19th - Saint Vincent And The Grenadines , did not appear
20th - Austria , ranked them 10th

Their ranking took into consideration way more information than my model however they did have sub-indexes like Economic development, health and natural environment which similarly aligned with my sub-indexes. For example this was Switzerland's breakdown:



CATEGORY	SCORE	RANK
Civic Health	74.5	#15
Culture & Tourism	73.6	#2
Economic Development	84.5	#1
Governance	87.0	#1
Health	84.0	#4
Infrastructure	78.8	#15
Natural Environment	65.3	#19
Opportunity	72.9	#2

Comparing my models results with the US news shows me that it performs decent enough. Despite missing data and limited data availability I think it's safe to say the model did a decent job at its predictions.

Closing Remarks

While I recognise that travel preferences will always remain partly subjective, this index shows that it is possible to build a structured framework capable of comparing countries using measurable indicators such as affordability, climate conditions, and safety risk.

Although the model is not perfect and remains limited by data availability, variable selection, and the simplification required when building composite indices, the final results were still able to produce rankings that aligned reasonably well with recognised travel destinations and external ranking systems. This suggests that the methodology I used was sufficiently robust to capture broad tourism suitability patterns across countries.

This project also reinforced the importance of data cleaning, feature engineering, dimensionality reduction, and critical interpretation when working with real-world datasets. More importantly, it highlighted to me that statistical outputs should never be accepted blindly, as unexpected rankings often reveal valuable insights about the assumptions and limitations built into the model itself.

In conclusion, I believe the final holiday index provides a useful proof of concept for how quantitative methods can support tourism analysis and destination comparison. With access to richer datasets and additional variables, the framework I developed in this project could be expanded further into a more advanced and realistic travel recommendation system.

References

- https://www.kaggle.com/datasets/mvieira101/global-cost-of-living – Global Cost of Living Dataset
- https://www.kaggle.com/datasets/nelgiriyewithana/global-weather-repository – Global Weather Repository Dataset
- EM-DAT: The International Disaster Database – Centre for Research on the Epidemiology of Disasters (CRED)
- https://www3.weforum.org/docs/WEF_Travel_and_Tourism_Development_Index_2024.pdf World Economic Forum Travel & Tourism Development Index 2024
- https://iits.business.gwu.edu/adventure-tourism-development-index George Washington University Adventure Tourism Development Index (ATDI)
- https://medium.com/@forhadhossan/why-weather-is-crucial-for-travel-a-practical-guide-2a1f428ea4be - Median Why Weather Is Crucial for Travel: A Practical Guide
- Statista - https://www.statista.com/chart/32515/influencing-factors-on-european-traveler-destination-decisions/
- Visit Saudi - https://www.visitsaudi.com/en/stories/climate-and-seasons
- CEC Network - https://www.studycanada.ca/english/climate.htm
- Rumbo Mundo - https://rumbomundo.com/weather-in-iceland-temperature-wind-precipitation/
- Weather Spark - https://weatherspark.com/h/y/104018/2023/Historical-Weather-during-2023-in-Riyadh-Saudi-Arabia
- Lost in Travel - https://lostintravel.blog/2024/01/20/every-country-i-visited-in-2023-ranked/
- U.S. News - https://www.usnews.com/news/best-countries/rankings